Free-body diagrams

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Draw only the external forces on one chosen object.

SUCCESS CRITERIA - I can...

1. Choose one object and list its external interactions.
2. Draw force arrows in physically correct directions.
3. Remove velocity and net force as extra interaction arrows.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “normal force” and “tension” to explain your answer. Sentence starters are optional.

normal force: a support force perpendicular to a contact surface

tension: the pulling force through a stretched rope or cable

weight: the gravitational force on an object

01 Warm-up: bring back an earlier idea

A fixed-mass cart has acceleration **3 m/s²** under **6 N** net force.
1. **Find acceleration** when net force becomes **10 N**.
2. Explain.

Word help: net force = the vector sum of all forces on one object

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Sigma F_x = ma_x$; $\Sigma F_y = ma_y$; $F_g = mg$.
Draw only forces on the chosen object. x and y name directions. Use g = **9.8 N/kg** unless another value is given.

Worked reasoning: A lamp hangs motionless from one vertical cable. With weight **12 N** down, the cable tension is **12 N** up. Only forces acting on the lamp belong in its diagram.

Your turn: 1. **Draw the lamp as a** labeled point with its two force arrows.
2. **Explain why its force** on the cable is not another arrow on this diagram.

Sentence starter: I think … . I can point to …, which shows …, so … .

Add and label force arrows.

Lamp: draw the two 12 N forces acting on it.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 A book at rest

1. Draw and **label the forces** on a book resting on a level table.
2. **Name both sources**.
3. **Use arrows** whose comparative lengths match rest.

Sentence starter: In my work, I used … to show … . I labeled it … .

Add and label force arrows.

The dot represents the object. Object: book

Marzano check - circle: 1 2 3

04 After the hand lets go

A box slides to the right on a rough floor after release.
1. **Draw its forces**.
2. Which way does friction act?

Sentence starter: In my work, I used … to show … . I labeled it … .

Add and label force arrows.

Box slides right after the hand lets go.

Marzano check - circle: 1 2 3

05 A falling ball

A ball falls down and is speeding up. Air pushes up.
1. **Draw weight** and air resistance.
2. Which arrow should be longer?

Sentence starter: In my work, I used … to show … . I labeled it … .

Add and label force arrows.

Falling ball speeds up; air resistance acts upward.

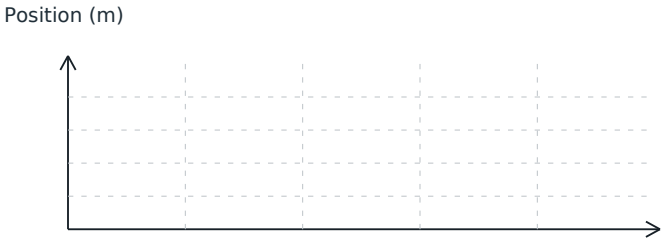
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart goes from **18 m** to **6 m** in **4 s** at steady velocity. **Find velocity** and **sketch its position-time graph**. **Label the diagram** with the given positions and any prediction.

Word help: velocity = rate of position change, with direction

Sentence starter: I predict … . I used … to make this prediction because … .



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A ball moves upward after leaving a hand. Ignore air resistance. **Draw its free-body diagram** and **explain why each arrow belongs** there.

Word help: justify = **support your answer** with a reason or evidence

Sentence starter: I think … . I can point to …, which shows …, so … .



Add and label force arrows.

Ball travels upward after leaving the hand.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Draw separate diagrams** for a hanging lamp at rest and the same lamp moving upward at steady speed.
2. What can a force diagram alone fail to tell you?

Word help: at rest = not moving in the chosen reference frame

Sentence starter: In my work, I used … to show … . I labeled it … .



Lamp at rest



Lamp moving up steadily

Marzano check - circle: 1 2 3